Topics: 14.1 Functions of Several Variables; 14.2 Limits and Continuity
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1. A scuba air tank is built from a right circular cylinder of radius x and height y, with two hemispherical end caps of radius x attached (one on each end of the cylinder).
   (a) Express the volume V of the tank as a function of x and y.

   The volume of the tank equals the volume of the cylinder plus the volume of the two hemispheres:
   $$V(x,y) = \pi x^2 y + 2 \cdot \frac{2}{3}\pi x^3 = \pi x^2 y + \frac{4}{3}\pi x^3.$$

   (b) Evaluate $V(8,5)$ and explain the meaning of this value.

   Evaluating $V(x,y)$ at $(8,5)$, we get that
   $$V(8,5) = \pi(8)^2(5) + \frac{4}{3}\pi(8)^3 = 320\pi + \frac{2048}{3}\pi = \frac{3008}{3}\pi \approx 3150.$$
   The volume of such a tank with radius $x = 8$ and height $y = 5$ is about 3150 (units^3).

2. Let g be the function defined by $g(x,y) = x\sin y + y\sin x$. Find the following.
   (a) $g(\pi, 0)$

   Substitute $x = \pi$ and $y = 0$:
   $$g(\pi,0) = \pi\sin 0 + 0\sin\pi = 0 + 0 = 0.$$

   (b) $g\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$

   Substitute $x = \frac{\pi}{2}$ and $y = \frac{\pi}{4}$:
   $$g\left(\frac{\pi}{2}, \frac{\pi}{4}\right) = \frac{\pi}{2}\sin\left(\frac{\pi}{4}\right) + \frac{\pi}{4}\sin\left(\frac{\pi}{2}\right).$$
   Using $\sin(\pi/4) = \frac{\sqrt{2}}{2}$ and $\sin(\pi/2) = 1$, we obtain
   $$g\left(\frac{\pi}{2}, \frac{\pi}{4}\right) = \frac{\pi}{2}\cdot\frac{\sqrt{2}}{2} + \frac{\pi}{4} = \frac{\pi\sqrt{2}}{4} + \frac{\pi}{4} = \frac{\pi}{4}(1+\sqrt{2}).$$

   (c) $g(0, y)$

   Substitute $x = 0$:
   $$g(0,y) = 0 \cdot \sin y + y\sin 0 = 0 + 0 = 0.$$

(d) $g(x, y+h)$

Replace y by $y+h$ in the formula for g:

$$g(x, y+h) = x\sin(y+h) + (y+h)\sin x.$$

3. Consider the function $F(x, y, z) = \sqrt{y} - \sqrt{x-2z}$.

(a) Evaluate $F(3, 4, 1)$.

Substitute $x = 3$, $y = 4$, and $z = 1$:

$$F(3, 4, 1) = \sqrt{4} - \sqrt{3-2(1)} = \sqrt{4} - \sqrt{1} = 2 - 1 = 1.$$

(b) Find the domain of F.

For $F(x, y, z)$ to be defined, both expressions under the square roots must be nonnegative:

$$y \geq 0 \qquad \text{and} \qquad x - 2z \geq 0.$$

Therefore the domain is

$$\{(x, y, z) \in \mathbb{R}^3 : y \geq 0 \text{ and } x - 2z \geq 0\}.$$

4. Find and sketch the domain of the function.

(a) $f(x, y) = \dfrac{\ln(2-x)}{1 - x^2 - y^2}$

For $f(x, y)$ to be defined, the logarithm must have a positive argument. Thus,

$$2 - x > 0 \quad \Longrightarrow \quad x < 2.$$

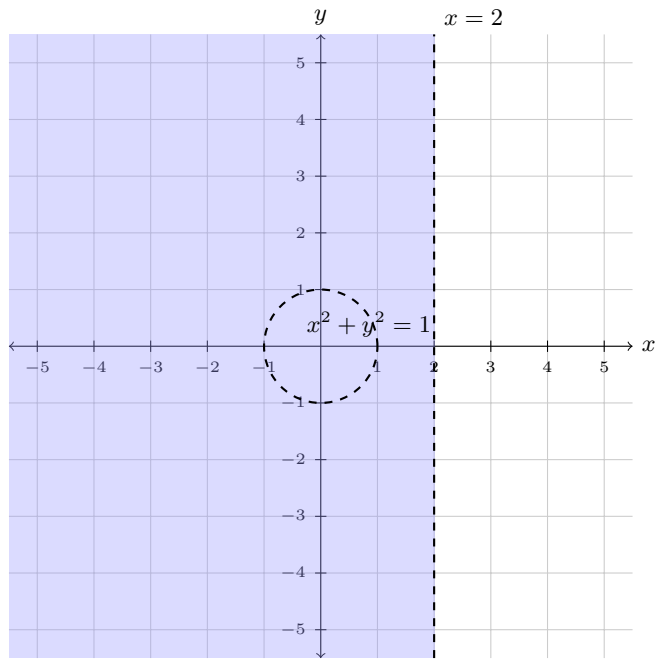
Also, the denominator cannot be zero, so we must exclude the points where

$$1 - x^2 - y^2 \neq 0 \quad \Longrightarrow \quad x^2 + y^2 \neq 1.$$

Therefore the domain is

$$D = \{(x, y) \in \mathbb{R}^2 : x < 2 \text{ and } x^2 + y^2 \neq 1\}.$$

Geometrically, this is the half-plane to the left of the line $x = 2$, with the circle $x^2 + y^2 = 1$ removed.



(b) $g(x, y) = \dfrac{\sqrt{x} - \sqrt{y}}{\sqrt{x^2 + y^2 - 4}}$

The function g is defined only when all radicals are real and the denominator is nonzero. Since $\sqrt{x}$ and $\sqrt{y}$ appear in the numerator, we must have

$$x \geq 0 \qquad \text{and} \qquad y \geq 0.$$

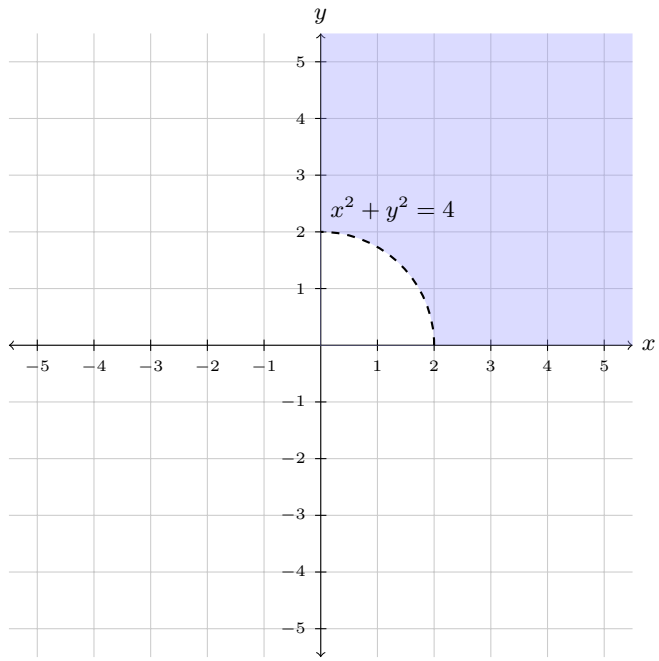
Also, the expression under the square root in the denominator must be positive (not merely non-negative, because it is in the denominator):

$$x^2 + y^2 - 4 > 0 \qquad \Longrightarrow \qquad x^2 + y^2 > 4.$$

Therefore the domain of g is

$$D = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0 \text{ and } x^2 + y^2 > 4\}.$$

Geometrically, this is the part of the first quadrant lying outside the circle $x^2 + y^2 = 4$.



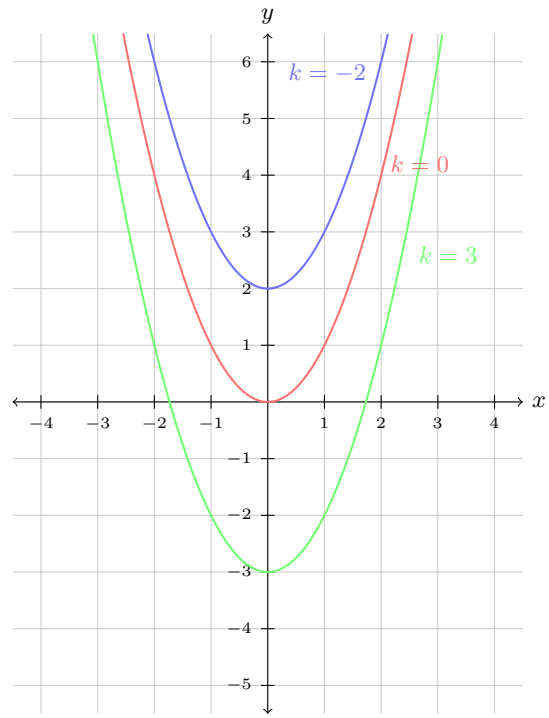
5. Sketch the level curves of the function for the given values of k.

   (a) $f(x, y) = x^2 - y$, for $k = -2, 0, 3$.

A level curve $f(x, y) = k$ satisfies

$$x^2 - y = k \quad \Longrightarrow \quad y = x^2 - k,$$

so the level curves are parabolas opening upward.

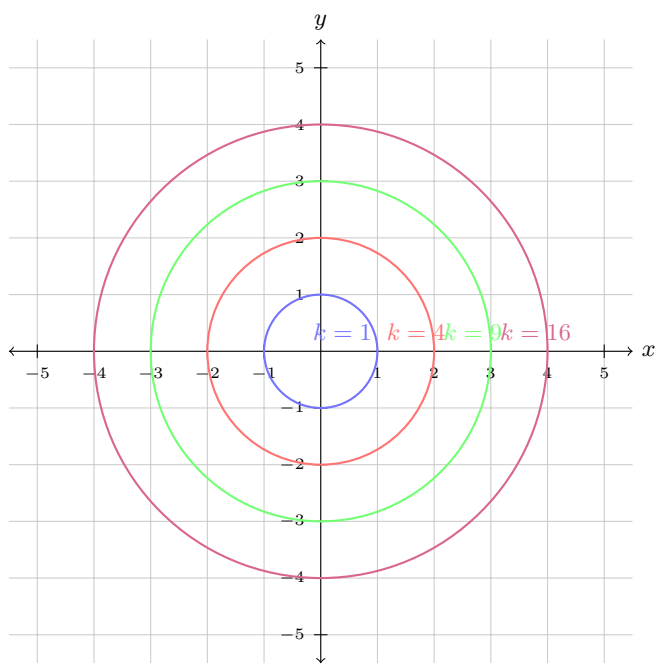


(b) $g(x, y) = x^2 + y^2$, for $k = 1, 4, 9, 16$.

A level curve $g(x, y) = k$ satisfies

$$x^2 + y^2 = k,$$

which is a circle centered at the origin with radius $\sqrt{k}$. Thus the requested level curves are circles of radii $1, 2, 3$, and 4 centered at the origin.



6. Evaluate the limit.

(a) $\lim_{(x,y)\to(3,2)} e^{\sqrt{2x-y}}$

We evaluate the limit by direct substitution:

$$\lim_{(x,y)\to(3,2)} e^{\sqrt{2x-y}} = e^{\sqrt{6-2}} = e^2.$$

(b) $\lim_{(x,y)\to(1,1)} \frac{x^2y^3 - x^3y^2}{x^2 - y^2}$

We evaluate the limit by factoring:

$$\lim_{(x,y)\to(1,1)} \frac{x^2y^3 - x^3y^2}{x^2 - y^2} = \lim_{(x,y)\to(1,1)} \frac{x^2y^2(y-x)}{(x-y)(x+y)} = \lim_{(x,y)\to(1,1)} -\frac{x^2y^2}{x+y} = -\frac{1}{2}.$$

(c) $\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$

We evaluate the limit by using the conjugate.

$$\begin{aligned}
\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}-1} &= \lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}-1} \cdot \frac{\sqrt{x^2+y^2+1}+1}{\sqrt{x^2+y^2+1}+1} \\
&= \lim_{(x,y)\to(0,0)} \frac{(x^2+y^2)(\sqrt{x^2+y^2+1}+1)}{(x^2+y^2+1)-1} \\
&= \lim_{(x,y)\to(0,0)} \frac{(x^2+y^2)(\sqrt{x^2+y^2+1}+1)}{x^2+y^2} \\
&= \lim_{(x,y)\to(0,0)} \sqrt{x^2+y^2+1}+1 \\
&= \sqrt{0^2+0^2+1}+1 \\
&= 2.
\end{aligned}$$

(d) $\lim_{(x,y,z)\to(1,-1,1)} \frac{2xz+3x+2yz+3y}{x+y}$

We evaluate the limit by factoring:

$$\begin{aligned}
\lim_{(x,y,z)\to(1,-1,1)} \frac{2xz+3x+2yz+3y}{x+y} &= \lim_{(x,y,z)\to(1,-1,1)} \frac{(x+y)(2z+3)}{x+y} \\
&= \lim_{(x,y,z)\to(1,-1,1)} 2z+3 \\
&= 2(1)+3 \\
&= 5.
\end{aligned}$$

7. Show that the limit does not exist.

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3x-2y}{4x^2-y^2}$

Let's approach $(0,0)$ along the line $y=x$:

$$\lim_{\substack{(x,y)\to(0,0)\\ y=x}} \frac{3x-2y}{4x^2-y^2} = \lim_{x\to 0} \frac{3x-2x}{4x^2-x^2} = \lim_{x\to 0} \frac{x}{3x^2} = \lim_{x\to 0} \frac{1}{3x}.$$

This one-variable limit does not exist (it is unbounded and approaches $+\infty$ from the right and $-\infty$ from the left). Therefore the two-variable limit does not exist as it does not exist along a path approaching $(0,0)$.

(b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+3y^2}$

Let's approach $(0,0)$ along the line $y=ax$, where a is a constant. Then

$$\lim_{\substack{(x,y)\to(0,0)\\ y=ax}} \frac{2xy}{x^2+3y^2} = \lim_{x\to 0} \frac{2x(ax)}{x^2+3(ax)^2} = \lim_{x\to 0} \frac{2ax^2}{(1+3a^2)x^2} = \frac{2a}{1+3a^2}.$$

Because the resulting value depends on a, different paths give different limits. For instance,

$$\lim_{\substack{(x,y)\to(0,0)\\ y=x}} \frac{2xy}{x^2+3y^2} = \frac{2}{1+3} = \frac{1}{2}, \qquad \lim_{\substack{(x,y)\to(0,0)\\ y=2x}} \frac{2xy}{x^2+3y^2} = \frac{4}{1+12} = \frac{4}{13}.$$

Since $\frac{1}{2} \neq \frac{4}{13}$, the limit does not exist.

8. Determine the set of points at which the function is continuous.

(a) $F(x,y) = \dfrac{1+x^2+y^2}{1-x^2-y^2}$

The numerator $1+x^2+y^2$ and the denominator $1-x^2-y^2$ are polynomials, so they are continuous everywhere. A quotient of continuous functions is continuous at every point where the denominator is nonzero. Thus F is continuous at all (x,y) satisfying

$$x^2+y^2 \neq 1.$$

Therefore F is continuous on the set

$$\{(x,y) \in \mathbb{R}^2 : x^2+y^2 \neq 1\}.$$

(b) $G(x,y) = \ln(1+x-y)$

The logarithm function is continuous on its domain, that is, wherever its argument is positive:

$$1+x-y > 0.$$

Therefore G is continuous on the set

$$\{(x,y) \in \mathbb{R}^2 : 1+x-y > 0\}.$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Functions of Two and Three Variables. A function of two variables is a rule that assigns to each point (x, y) in a subset of $\mathbb{R}^2$ a real number. A function of three variables is a rule that assigns to each point (x, y, z) in a subset of $\mathbb{R}^3$ a real number.
Unless otherwise specified, by the **domain** of $f(x, y)$, we mean the set of all points (x, y) for which the expression for f produces a real number (similarly for functions of three variables).

Level Curves and Level Surfaces. The **level curve** of $f(x, y)$ corresponding to the value k is the curve given by

$$f(x, y) = k.$$

For functions of three variables, the analogous objects are **level surfaces**, given by

$$g(x, y, z) = k.$$

Limits. If $f(x, y)$ can be made arbitrarily close to L by taking (x, y) sufficiently close to (a, b) (but not necessarily equal to (a, b)), then we write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L.$$

On the other hand, if $f(x, y)$ does not approach a single value as (x, y) approaches (a, b), then the above limit does not exist. Analogous definitions apply for functions of three variables.

Continuity. A function f is **continuous** at (a, b) if

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b).$$

Functions built from polynomials, trigonometric functions, exponentials, rational functions (where denominators are nonzero), square roots (argument ≥ 0), and logarithms (argument > 0) are continuous on their domains. The same is true for functions of three variables.

Suggested Textbook Problems
Section 14.1: 1-56, 61-72
Section 14.2: 1-50